Reader: Now all becomes very simple. We know that the sequence (d_n) is infinitesimal. It means that for any $\varepsilon' > 0$ we can find N such that for all $n > N$ $|d_n| < \varepsilon'$.

- $\boxed{\text{For } \varepsilon', \text{ I select } \frac{\varepsilon}{M}}$ $\rightarrow$ $\forall \varepsilon'$, does $\exists \varepsilon$ s.t. $\varepsilon' = \frac{\varepsilon}{M}$? Is it necessarily true?
- Then, for $n > N$ we have $|y_n d_n| = |y_n||d_n| \leq M|d_n| < M \cdot \frac{\varepsilon}{M} = \varepsilon$ Does this really cover $\forall \varepsilon \in \mathbb{R}$?

This completes the proof.

Author: Excellent. Now, making use of this theorem, it is very easy to prove another

Theorem: A sequence $(y_n z_n)$ is convergent to ab if sequences (y_n) and (z_n) are convergent to a and b, respectively.

Suppose $y_n = a + d_n$ and $z_n = b + \beta_n$. Suppose also that the sequences (d_n) and (β_n) are infinitesimal. Then we can write:

$$\boxed{y_n z_n = ab + \gamma_n, \text{ where } \gamma_n = a\beta_n + d_n b + d_n \beta_n}$$

Making use of the theorem we have just proved, we conclude that the sequences (bd_n), $(a\beta_n)$ and $(d_n\beta_n)$ are infinitesimal.

Reader: But what justifies your conclusion about the sequence $(d_n\beta_n)$?

Author: Because any convergent sequence (regardless of whether it is infinitesimal or not) is bounded.

From the theorem on the sum of convergent sequences we infer that the sequence (γ_n) is infinitesimal, which immediately yields

$$\boxed{\lim_{n \to \infty} (y_n z_n) = ab}$$

. This completes the proof. □

Reader: Perhaps we should also analyze inverse variants in which the sequence $(y_n z_n)$ is convergent. What can be said in this case about the sequences (y_n) and (z_n)?